
Highly-Parallel Algorithms for Non-Smooth Non-Convex Optimization

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Abstract

Recent progress in nonsmooth nonconvex optimization established non-asymptotic convergence to (δ, ϵ) -Goldstein stationary points with zeroth- or first-order oracles. In this paper, we study the extensions of these results in the parallel setting. We propose a algorithm called Optimistic Nonconvex Gradient (NOG) that can find a (δ, ϵ) -Goldstein stationary point in $\mathcal{O}(d^{1/3}\delta^{-1}\epsilon^{-5/3})$ rounds of interactions of the oracles, which strictly improve the one of $\mathcal{O}(\delta^{-1}\epsilon^{-3})$ given by prior arts when $d = \mathcal{O}(\epsilon^{-4})$. We further implement our method in the distributed and quantum optimization settings, obtaining efficient distributed algorithms in both communication and computation, and fast quantum algorithms in quantum oracle query complexities.

1 Introduction

This paper studies the stochastic optimization problem of the form

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) \triangleq \mathbb{E}_{\xi \in \Xi} [F(\mathbf{x}; \xi)], \quad (1)$$

where the stochastic component $F(\mathbf{x}; \xi)$, indexed by random variable ξ defined on the data distribution Ξ , is possibly nonconvex and nonsmooth. This nonsmooth nonconvex optimization covers a lot of applications in statistics [20, 54], economics [19, 48], and deep learning [3, 14, 33, 53].

For nonsmooth nonconvex optimization, Zhang et al. [53] showed traditional ϵ -stationary point cannot be obtained in finite time and proposed the Stochastic Interpolated Normalized Gradient Descent (SINGD) that can provably find a (δ, ϵ) -Goldstein stationary point [25] in $\mathcal{O}(\delta^{-1}\epsilon^{-4})$ stochastic first-order oracle (SFO) complexity. Cutkosky et al. [12] proposed optimal first-order methods with $\mathcal{O}(\delta^{-1}\epsilon^{-3})$ SFO complexity via a novel Online-to-Nonconvex Conversion (O2NC). Kornowski and Shamir [32] showed a zeroth-order implementation of O2NC with two-point gradient estimator [18, 46] achieves a $\mathcal{O}(d\delta^{-1}\epsilon^{-3})$ stochastic zeroth-order oracle (SZO) complexity.

Although optimal first-order [12] and optimal dimension-dependent zeroth-order methods [32] have been proposed, these methods are studied under the sequential setting. With the growing size of training datasets, modern training often relies on parallel computing resources like GPUs and TPUs. To capture this scenario, we consider the parallel setting, where the algorithm can simultaneously submit multiple queries $\{(\mathbf{x}_1, \xi_1), \dots, (\mathbf{x}_Q, \xi_Q)\}$ at once, and then the zeroth- or first-order oracles return the function values or gradients at these points to the algorithm. Following the work by Nemirovski [39], the complexity of a parallel algorithm is measured by both *depth* (the number of rounds the algorithm interact with the oracles) and *work* (the total number of queries, *i.e.*, $Q \times \text{depth}$). This work tries to answer the following fundamental question:

Can we develop fast parallel algorithms for nonsmooth nonconvex optimization?

Oracle	Method	Depth	Work	Reference
0th	GFM	$\mathcal{O}(d^{1/2}\delta^{-1}\epsilon^{-2})$	$\mathcal{O}(d^{3/2}\delta^{-1}\epsilon^{-4})$	[34]
	GFM+	$\mathcal{O}(d^{1/2}\delta^{-1}\epsilon^{-2})$	$\mathcal{O}(d^{3/2}\delta^{-1}\epsilon^{-3})$	[7]
	O2NC	$\mathcal{O}(\delta^{-1}\epsilon^{-3})$	$\mathcal{O}(d\delta^{-1}\epsilon^{-3})$	[32]
	NOG(ours)	$\mathcal{O}(d^{1/3}\delta^{-1}\epsilon^{-5/3})$	$\mathcal{O}(d\delta^{-1}\epsilon^{-3})$	Theorem 4.2
1st	INGD	$\mathcal{O}(\delta^{-1}\epsilon^{-4})$	$\mathcal{O}(\delta^{-1}\epsilon^{-4})$	[15, 50, 53]
	O2NC	$\mathcal{O}(\delta^{-1}\epsilon^{-3})$	$\mathcal{O}(\delta^{-1}\epsilon^{-3})$	[12]
	NOG(ours)	$\mathcal{O}(d^{1/3}\delta^{-1}\epsilon^{-5/3})$	$\mathcal{O}(\delta^{-1}\epsilon^{-3})$	Theorem 4.3

Table 1: Comparison of depth (rounds of interactions) and work (total query complexities) for **classical** algorithms for finding (δ, ϵ) stationary points of nonsmooth nonconvex problems.

Although there have been many works studying the complexity of parallel algorithms for convex problems [1, 4, 5, 16, 17, 23, 24, 27, 45, 47] or smooth nonconvex problems [52, 56], the power of parallelism in nonsmooth nonconvex optimization is less explored. To fill the gap, this work extends the randomized smoothing techniques in Duchi et al. [17] for nonsmooth convex problems to nonsmooth nonconvex problems, and proposes a novel Nonconvex Optimistic Gradient (NOG) method that achieves a low depth of $\mathcal{O}(d^{1/3}\delta^{-1}\epsilon^{-5/3})$ for finding a (δ, ϵ) -Goldstein stationary point. The depth of our method improves the $\mathcal{O}(\delta^{-1}\epsilon^{-3})$ of O2NC [12, 32] and the $\mathcal{O}(\sqrt{d}\delta^{-1}\epsilon^{-2})$ of naive gradient descent with randomized smoothing [7, 29, 34], and the total work of our method aligns with the state-of-the-art ones obtained by O2NC [12, 32] when using either first- or zeroth-order oracles, *i.e.*, $\mathcal{O}(\delta^{-1}\epsilon^{-3})$ SFO complexity and $\mathcal{O}(d\delta^{-1}\epsilon^{-3})$ SZO complexity. We compare the theoretical guarantee of the proposed NOG with previous ones in Table 1.

Our algorithm is well-suited for various scenarios requiring high parallelism. A natural application is distributed optimization, where multiple parallel workers collaborate to perform optimization tasks. The complexity of a distributed algorithm is typically measured by communication and computation costs; specifically, communication is determined by the depth of the algorithm, while computation is equivalent to its total work. Consequently, our algorithm can be efficiently implemented in distributed settings, significantly reducing the communication complexity compared to prior distributed algorithms [8, 35, 43]. We summarize our results for distributed algorithms in Table 2.

2 Related Works

2.1 First-order nonsmooth nonconvex optimization

Works before 2020 typically can only obtain asymptotic convergence of first-order methods to ϵ -stationary points [2, 3, 14, 30, 38] for nonsmooth nonconvex optimization. Around 2020, various hardness results were established, showing that traditional ϵ -stationary points can be intractable in finite time for any algorithms [29, 31, 49, 53]. Zhang et al. [53] introduced the notion of Goldstein stationary points as a tractable goal for non-asymptotic convergence rates and proposed the Stochastic Interpolated Normalized Gradient Descent (SINGD) algorithm that can find a (δ, ϵ) -Goldstein stationary point in the $\mathcal{O}(\delta^{-1}\epsilon^{-4})$ SFO complexity for Hadamard semi-differentiable functions. Davis et al. [15], Tian et al. [50] leveraged the Rademacher's theorem and proposed the perturbed INGD algorithm that achieves the same complexity using standard stochastic first-order oracles. Additionally, Davis et al. [15] showed that finding the minimal norm of Goldstein subdifferential via a cutting plane method in the INGD algorithm [53] achieves the dimension-dependent first-order oracle complexity of $\mathcal{O}(d\delta^{-1}\epsilon^{-2})$ in the deterministic setting. Remarkably, Cutkosky et al. [12] proposed an Online-to-Nonconvex Conversion (O2NC) framework to reduce finding Goldstein stationary points to online learning, which achieves the SFO complexity of $\mathcal{O}(\delta^{-1}\epsilon^{-3})$. Moreover, Cutkosky et al. [12] also proposed a matching lower bound of $\Omega(\delta^{-1}\epsilon^{-3})$ for stochastic first-order algorithms, showing the complexity achieved by O2NC is indeed optimal.

69 2.2 Zeroth-order nonsmooth nonconvex optimization

70 In terms of zeroth-order methods, Lin et al. [34] first introduced randomized smoothing [17] to
 71 nonsmooth nonconvex optimization and proposed the Gradient Free Method (GFM) that finds a
 72 (δ, ϵ) -Goldstein stationary point in $\mathcal{O}(d^{3/2}\delta^{-1}\epsilon^{-4})$ SZO complexity. Chen et al. [7] proposed a faster
 73 method called GFM+ that achieves a lower SZO complexity of $\mathcal{O}(d^{3/2}\delta^{-1}\epsilon^{-3})$ via variance reduction
 74 [11, 21, 55]. Kornowski and Shamir [32] showed that using a zeroth-order gradient estimator [46] in
 75 O2NC [12] gives a better algorithm with $\mathcal{O}(d\delta^{-1}\epsilon^{-3})$ SZO complexity. Liu et al. [36] proposed a
 76 quantum implementation of GFM+ to achieve the $\mathcal{O}(d^{3/2}\delta^{-1}\epsilon^{-7/3})$ QSZO complexity, which can
 77 outperform classical algorithms in the ϵ dependency.

78 3 Preliminaries

79 **Notations.** Throughout this paper, we use $\|\cdot\|$ to denote both the Euclidean norm of a vector.
 80 We let $\mathbb{B}_D^d(c)$ denote the ball $\{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x} - c\| \leq D\}$, and denote $\mathbb{B}^d = \mathbb{B}_1^d(\mathbf{0})$ and $\mathbb{S}^{d-1}$ to be
 81 the d -dimensional unit ball and unit sphere, respectively. For a set S , we let $\Pi_S(\cdot)$ to denote the
 82 projection operator onto it. We also let $\text{Unif}(S)$ be the uniform distribution over the set S . For
 83 functions $g, h : S \rightarrow [0, \infty)$ we write $g = \mathcal{O}(h)$ or $h = \Omega(g)$ equivalently if there exists $c > 0$ such
 84 that $g(s) \leq ch(s)$ for every $s \in S$. We also write $h = \Theta(g)$ if $h = \mathcal{O}(g)$ and $h = \Omega(g)$, and write
 85 $\tilde{\mathcal{O}}(\cdot)$ to hide the logarithmic factors in $\mathcal{O}(\cdot)$.

86 3.1 Nonsmooth Nonconvex Optimization

87 Following the literature [7, 12, 17, 32, 34, 36, 50, 53], we make the following assumptions on the
 88 objective function for zeroth- and first-order nonsmooth nonconvex optimization.

89 **Assumption 3.1.** We assume the objective function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies:

- 90 (a) $f^* = \inf_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) > -\infty$ and $f(\mathbf{0}) - f^* \leq \gamma$.
- 91 (b) The stochastic component $F(\cdot; \xi)$ of the objective $f(\cdot)$ satisfies $\mathbb{E}_{\xi \in \Xi}[F(\mathbf{x}; \xi)] = f(\mathbf{x})$
 92 and $F(\cdot; \xi)$ is $L(\xi)$ -Lipschitz with $\mathbb{E}_{\xi \sim \Xi}[L(\xi)^2] \leq L^2$.

93 Note that the second assumption implies that the objective $f(\cdot)$ is L -Lipschitz continuous, and hence
 94 the Rademacher's theorem indicates that f is differentiable almost everywhere. This allows us to
 95 define the Clarke subdifferentiable [9] in the following way.

96 **Definition 3.1** (Clarke subdifferential). The Clarke subdifferential of a Lipschitz function f at a point
 97 $\mathbf{x}$ is defined by $\partial f(\mathbf{x}) := \text{conv} \{\mathbf{g} : \mathbf{g} = \lim_{\mathbf{x}_k \rightarrow \mathbf{x}} \nabla f(\mathbf{x}_k)\}$.

98 The above Clarke subdifferentiable is a natural generalization of gradients for smooth objectives.
 99 However, it was proved that finding a point with a small Clarke subdifferentiable ([53, Theorem 2]),
 100 or even approaching a neighborhood of such points ([31, Theorem 1]), can be intractable in finite
 101 time. To address this issue, Zhang et al. [53] proposed the following (δ, ϵ) -Goldstein stationarity.

102 **Definition 3.2** (Approximate Goldstein stationary point). Given a Lipschitz function f , we say
 103 the point $\mathbf{x}$ is a (δ, ϵ) -Goldstein stationary point of f if it holds that $\text{dist}(\mathbf{0}, \partial_\delta f(\mathbf{x})) \leq \epsilon$, where
 104 $\partial_\delta f(\mathbf{x}) := \text{conv} \left\{ \bigcup_{\mathbf{y} \in \mathbb{B}_\delta^d(\mathbf{x})} \partial f(\mathbf{y}) \right\}$ is the Goldstein subdifferentiable introduced by Goldstein [25].

105 We consider two setups: (1) zeroth-order setup, where the algorithm has access to the function
 106 value information $F(\mathbf{x}; \xi)$ at the query point $\mathbf{x}$ and random index ξ ; (2) first-order setups, where the
 107 algorithm has access to the gradient information $\partial F(\mathbf{x})$. We formally measure the complexities of
 108 algorithms by the number of oracles calls in the respective setups.

109 **Definition 3.3** (Oracle complexity). (1) For a zeroth-order algorithm, we define its stochastic zeroth-
 110 order oracles (SZO) complexity as the minimal number of queries $F(\mathbf{x}; \xi)$ to find a point $\hat{\mathbf{x}}$ such that
 111 $\mathbb{E}[\text{dist}(\mathbf{0}, \partial_\delta f(\hat{\mathbf{x}}))] \leq \epsilon$. (2) Similarly, for a first-order algorithm, we define its stochastic first-order
 112 oracle (SFO) complexity as the minimal number of queries $\partial F(\mathbf{x}; \xi)$ to find a point $\hat{\mathbf{x}}$ such that
 113 $\mathbb{E}[\text{dist}(\mathbf{0}, \partial_\delta f(\hat{\mathbf{x}}))] \leq \epsilon$.

Algorithm 1 $\text{NOG}(K, M, D)$

Require: a (δ, σ) -SFO $\mathcal{O}$ as per Definition 3.5.

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1:  $\mathbf{x}_0 = \mathbf{y}_0 = \mathbf{y}_{-1} = \mathbf{\Delta}_0 = \mathbf{0}$  and  $T = KM$ 
2: for  $t = 1, \dots, T$  do
3:    $\mathbf{\Delta}_t = \Pi_{\mathbb{B}_D^d(\mathbf{0})}(\mathbf{\Delta}_{t-1} - 2\eta\mathcal{O}(\mathbf{y}_{t-1}) + \eta\mathcal{O}(\mathbf{y}_{t-2}))$ .
4:   Sample  $s_t \sim \text{Unif}[0, 1]$ 
5:    $\mathbf{y}_t = \mathbf{x}_{t-1} + s_t\mathbf{\Delta}_t$ ,  $\mathbf{x}_t = \mathbf{x}_{t-1} + \mathbf{\Delta}_t$ 
6: end for
7: for  $k = 1, \dots, K$  do
8:    $\bar{\mathbf{y}}_k = \frac{1}{M} \sum_{m=1}^M \mathbf{y}_{(k-1)M+m}$ 
9: end for
10:  $k_{\text{out}} \sim \text{Unif}\{1, \dots, K\}$  return  $\bar{\mathbf{y}}_{\text{out}}$ 
```

114 3.2 Randomized Smoothing

115 A tool used in our paper is the randomized smoothing [17, 51], which smoothed a nonconvex objective
116 via adding random perturbations from a density function p . We leverage the randomized smoothing
117 with p being the uniform density on the ℓ_2 -ball with radius δ , which leads to the following definition.

118 **Definition 3.4** (δ -smoothed surrogate). *For a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, we denote its δ -smoothed*
119 *surrogate as*

$$f_\delta(\mathbf{x}) := \mathbb{E}_{\mathbf{u} \sim \text{Unif}(\mathbb{B}^d)}[f(\mathbf{x} + \delta\mathbf{u})].$$

120 We summarize the known properties of this surrogate [17, 32, 34, 51] in the following.

121 **Proposition 3.1.** *If f satisfies Assumption 3.1, then its smoothed surrogate f_δ satisfies:*

- 122 (a) $|f_\delta(\cdot) - f(\cdot)| \leq \delta L$.
- 123 (b) $f_\delta(\cdot)$ is L -Lipschitz.
- 124 (c) $\partial_\delta f_\delta(\cdot) \in \partial_{2\delta} f(\cdot)$.
- 125 (d) $f_\delta(\cdot)$ is differentiable and has $\mathcal{O}(\sqrt{d}L/\delta)$ -Lipschitz gradients.

126 Next, we introduce the concept of (δ, σ) -SFO, which is essentially the SFO of the smoothed surro-
127 gate f_δ with variance σ^2 . It will serve as an intermediate oracle in our analyses.

128 **Definition 3.5.** *For a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, the (δ, σ) -stochastic first-order oracle (SFO) $\mathcal{O} : \mathbb{R}^d \rightarrow$*
129 *$\mathbb{R}^d$ returns a random variable $\mathcal{O}(\mathbf{x})$ such that (a) $\mathbb{E}[\mathcal{O}(\mathbf{x})] = \nabla f_\delta(\mathbf{x})$ and (b) $\text{Var}[\mathcal{O}(\mathbf{x})] \leq \sigma^2$.*

130 4 Optimistic Nonconvex Gradient Method

131 In this section, we propose a meta-algorithm called Optimistic Nonconvex Gradient (NOG) for highly-
132 parallel nonsmooth nonconvex optimization, whose complete procedure is presented in Algorithm 1.
133 The remainder of this section is organized as follows. First, in Section 4.1, we showed that NOG is
134 equivalent to applying the standard optimistic gradient method [26, 42] to solving the online learning
135 problem in [12]; Then, in Section 4.2 provides a simple complexity analysis of the NOG method using
136 this equivalent viewpoint, showing that it can find a (δ, ϵ) -Goldstein stationary point given access
137 to a (δ, σ) -SFO of f . Finally, in Section 4.3, we instantiate the (δ, σ) -SFO using SFO/SZO of f
138 with known estimators, resulting in highly-parallel zeroth- and first-order methods for nonsmooth
139 nonconvex optimization. .

140 4.1 Design of the Algorithm

141 To develop our algorithms, we first follow Cutkosky et al. [12] to convert the task of finding a
142 (δ, ϵ) -Goldstein stationary point to solving an online direction learning problem. This reduction
143 is done via the online-to-nonconvex conversion (O2NC) described in Algorithm 2. We apply the
144 O2NC [12] to the smoothed surrogate f_δ (Definition 3.4) with the (δ, σ) -SFO (Definition 3.5) to

145 obtain the following result. Note that f_δ is everywhere differentiable, and automatically satisfies the
 146 “well-bahavedness” [12, Definition 1] as required.

147 **Lemma 4.1** ([12, Theorem 8] applied on f_δ). *Running Algorithm 2 with $D = \delta/M$ and $T = KM$,
 148 then the algorithm ensures that (a) For all k and m , $\|\mathbf{y}_k - \mathbf{y}_{(k-1)M+m}\| \leq \delta$. (b) We have*

$$\frac{1}{K} \mathbb{E} \left\| \frac{1}{M} \sum_{t=(k-1)M+1}^{kM} \nabla f_\delta(\mathbf{y}_t) \right\| \leq \frac{f_\delta(\mathbf{x}_0) - f_\delta^*}{KDM} + \frac{\mathbb{E}[\text{Reg}_{T,K}]}{KDM} + \frac{\sigma}{\sqrt{M}}. \quad (2)$$

149 where the K -shifting regret is defined as

$$\text{Reg}_{T,K} := \sup_{\mathbf{u}_1, \dots, \mathbf{u}_K \in \mathbb{B}_D^d(\mathbf{0})} \sum_{k=1}^K \sum_{t=(k-1)M+1}^{kM} \langle \mathbf{O}(\mathbf{y}_t), \Delta_t - \mathbf{u}_k \rangle. \quad (3)$$

150 From Lemma 4.1, a small expected regret implies the right-hand side of inequality (2) is also small,
 151 which gives a certificate of an approximate Goldstein stationary point. In the literature of online
 152 learning, a well-studied algorithm for regret minimization is the optimistic gradient (OG) method
 153 [12, 13, 26, 28, 41, 42]. With the naive hint $\mathbf{h}_t = \mathbf{O}(\mathbf{y}_{t-1})$, this method has the following update:

$$\Delta_t = \Pi_{\mathbb{B}_D^d(\mathbf{0})}(\Delta_{t-1} - 2\eta \mathbf{O}(\mathbf{y}_{t-1}) + \eta \mathbf{O}(\mathbf{y}_{t-2})), \quad (4)$$

154 and is known to have the following guarantee.

155 **Lemma 4.2** (Patitucci et al. [41, Lemma 3.1] with $\mathbf{h}_t = \mathbf{O}(\mathbf{y}_{t-1})$). *For any vectors $\mathbf{u}_1, \dots, \mathbf{u}_K \in$
 156 $\mathbb{B}_D^d(\mathbf{0})$, the update in Eq. (4) ensures the regret bound of*

$$\mathbb{E}[\text{Reg}_{T,K}] \leq \frac{4KD^2}{\eta} + \frac{5\eta}{2} \sum_{t=1}^{KM} \mathbb{E} \|\mathbf{O}(\mathbf{y}_t) - \mathbf{O}(\mathbf{y}_{t-1})\|^2.$$

157 Finally, we obtain the proposed method in Algorithm 1 by instantiating the online learning algorithm A
 158 in Algorithm 2 with the OG method (4), and specifying the output point according to the left-hand
 159 side of inequality (2).

Algorithm 2 O2NC on $f_\delta(K, M, D)$

Input: a (δ, σ) -SFO $\mathbf{O}$ as per Definition 3.5 and an online learning algorithm A.

- 1: $\mathbf{x}_0 = \mathbf{y}_0 = \mathbf{y}_{-1} = \Delta_0 = \mathbf{0}$ and $T = KM$
 - 2: **for** $t = 1, \dots, T$
 - 3: Send $\mathbf{O}(\mathbf{y}_{t-1})$ to A and receive $\Delta_t \in \mathbb{B}_D^d(\mathbf{0})$ from A
 - 4: Sample $s_t \sim \text{Unif}[0, 1]$
 - 5: $\mathbf{y}_t = \mathbf{x}_{t-1} + s_t \Delta_t$, $\mathbf{x}_t = \mathbf{x}_{t-1} + \Delta_t$
 - 6: **end for**
-

160 4.2 Complexity Analysis

161 This section presents the complexity analysis of Algorithm 1. To proceed from Lemma 4.3, we
 162 leverage the smoothness of f_δ to give a tighter upper bound of bound $\mathbb{E} \|\mathbf{O}(\mathbf{y}_t) - \mathbf{O}(\mathbf{y}_{t-1})\|^2$, leading
 163 to the following result.

164 **Lemma 4.3.** *Let $\mathbf{O}$ be a (δ, σ) -SFO of a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ that satisfies Assumption 3.1. For any
 165 vectors $\mathbf{u}_1, \dots, \mathbf{u}_K \in \mathbb{B}_D^d(\mathbf{0})$, the update in Eq. (4) ensures the regret bound of*

$$\mathbb{E}[\text{Reg}_{T,K}] \leq \frac{4KD^2}{\eta} + 15\eta KM \left(\frac{9c^2 d L^2 D^2}{2\delta^2} + \sigma^2 \right).$$

166 Finally, by combining the O2NC guarantee by Lemma 4.1 and appropriately choosing the hyperpa-
 167 rameters, we obtain the following theorem for the proposed NOG method.

Theorem 4.1. Let $\mathbf{O}$ be a (δ, σ) -SFO of a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ that satisfies Assumption 3.1. Setting the parameters in Algorithm 1 as

$$\eta = \Theta \left(\frac{\delta}{L\sqrt{dM}} \wedge \frac{D}{\sigma\sqrt{M}} \right) \quad \text{and} \quad M = \Theta \left(\left(\frac{\sqrt{d}L\delta T}{\gamma + \delta L} \right)^{2/5} \vee \left(\frac{\sigma\delta T}{\gamma + \delta L} \right)^{2/3} \right).$$

Then the algorithm can find a $(2\delta, \epsilon)$ -Goldstein stationary point of f in iterations no more than

$$T = \mathcal{O} \left(\frac{\gamma + \delta L}{\delta} \left(\frac{d^{1/3}L^{2/3}}{\epsilon^{5/3}} + \frac{\sigma^2}{\epsilon^3} \right) \right).$$

In the next subsection, we further specify the (δ, σ) -SFO required by Theorem 4.1 using the standard gradient estimators for f_δ [17, 18, 32, 34, 43, 46, 51], which results in highly-parallel zeroth- and first-order methods for nonsmooth nonconvex optimization.

4.3 Highly-Parallel Zeroth- and First-Order Methods

In the zeroth-order setup, it is known that the following mini-batch two-point estimator [18, 32, 34, 43, 46] gives a (δ, σ) -SFO with $B = \mathcal{O}(dL^2/\sigma^2)$ SZO calls.

Proposition 4.1 (Zeroth-order estimator). Under Assumption 3.1, a (δ, σ) -SFO of function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ can be implemented by setting $\mathbf{O}(\mathbf{x}) = \frac{1}{B} \sum_{j=1}^B \mathbf{O}_j(\mathbf{x})$ with batch size $B = \mathcal{O}(dL^2/\sigma^2)$ and

$$\mathbf{O}_j(\mathbf{x}) = \frac{d}{2\delta} (F(\mathbf{x} + \delta \mathbf{v}; \xi) - F(\mathbf{x} - \delta \mathbf{v}; \xi) \mathbf{v}), \quad \mathbf{v} \sim \text{Unif}(\mathbb{S}^{d-1}), \quad \xi \sim \Xi.$$

Then, substituting into Theorem 4.1, we obtain the following complexity of zeroth-order implementation of the NOG method.

Theorem 4.2 (Zeroth-order NOG). Under Assumption 3.1, Algorithm 1 with the parameters given by Theorem 4.1 and the (δ, σ) -SFO given by Proposition 4.1 with $\sigma = \mathcal{O}(d^{1/6}L^{1/3}\epsilon^{2/3})$ can find a $(2\delta, \epsilon)$ -Goldstein stationary point of f in depth of $\mathcal{O}(d^{1/3}(\gamma + \delta L)L^{2/3}\delta^{-1}\epsilon^{-5/3})$ and SZO complexity of $\mathcal{O}(d(\gamma + \delta L)L^2\delta^{-1}\epsilon^{-3})$.

In the first-order setup, the following estimator [17, 43, 51] gives an unbiased estimate of $\nabla f_\delta(\mathbf{x})$, which is thus a (δ, σ) -SFO when choosing the batch size $B = \mathcal{O}(L^2/\sigma^2)$.

Proposition 4.2 (First-order estimator). Under Assumption 3.1, a (δ, σ) -SFO of function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ can be implemented by setting $\mathbf{O}(\mathbf{x}) = \frac{1}{B} \sum_{j=1}^B \mathbf{O}_j(\mathbf{x})$ with batch size $B = \mathcal{O}(L^2/\sigma^2)$ and

$$\mathbf{O}_j(\mathbf{x}) = \nabla F(\mathbf{x} + \delta \mathbf{u}; \xi), \quad \text{where } \mathbf{u} \sim \text{Unif}(\mathbb{B}^d), \quad \xi \sim \Xi.$$

We have the following theorem for the first-order implementation of the NOG method. Our result recovers the optimal SFO complexity for finding (δ, ϵ) -Goldstein stationary points [12], and, same as the zeroth-order implementation, improves the depth of previous methods [12] by a factor of $\mathcal{O}(\epsilon^{-4/3}/d^{1/3})$ when $d = \mathcal{O}(\epsilon^{-4})$.

Theorem 4.3 (First-order NOG). Under Assumption 3.1, Algorithm 1 with the parameters given by Theorem 4.1 and the (δ, σ) -SFO given by Proposition 4.2 with $\sigma = \mathcal{O}(d^{1/6}L^{1/3}\epsilon^{2/3})$ can find a $(2\delta, \epsilon)$ -Goldstein stationary point of f in depth of $\mathcal{O}(d^{1/3}(\gamma + \delta L)L^{2/3}\delta^{-1}\epsilon^{-5/3})$ and SFO complexity of $\mathcal{O}((\gamma + \delta L)L^2\delta^{-1}\epsilon^{-3})$.

Theorem 5.1 and 5.1 show that our proposed method has lower depth than existing ones for nonsmooth nonconvex optimization. In the next sections, we show that this also gives more efficient algorithms in various settings that favor parallel computing.

5 Applications in Distributed Optimization

In this section, we provide an application of the NOG method in distributed optimization over a strongly connected directed graph of m nodes [37, 44, 45], each having access to a local nonconvex

203 nonsmooth function $f_i(\mathbf{x}) \triangleq \mathbb{E}_{\xi \in \Xi_i} [F_i(\mathbf{x})]$. The goal is to minimize the average of all the m local
 204 functions:

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) \triangleq \frac{1}{m} \sum_{i=1}^m f_i(\mathbf{x}), \quad \text{where} \quad f_i(\mathbf{x}) \triangleq \mathbb{E}_{\xi \in \Xi_i} [F_i(\mathbf{x})]. \quad (5)$$

205 By imposing the Lipschitz conditions on each local objective $f_i(\mathbf{x})$, we extend Assumption 3.1 for
 206 single-agent optimization to the distributed setting as follows.

207 **Assumption 5.1.** *In the distributed setting, we assume that:*

- 208 (a) *The global objective $f(\cdot)$ satisfies $f^* = \inf_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) > -\infty$ and $f(\mathbf{0}) - f^* \leq \gamma$.*
- 209 (b) *For each local objective $f_i(\cdot)$, the stochastic component $F_i(\cdot; \xi)$ satisfies $\mathbb{E}_{\xi \in \Xi} [F_i(\mathbf{x}; \xi)] =$
 210 $f_i(\mathbf{x})$ and $F_i(\cdot; \xi)$ is $L_i(\xi)$ -Lipschitz with $\mathbb{E}_{\xi \sim \Xi_i} [L_i(\xi)^2] \leq L^2$.*

211 Next, we formally define the distributed algorithm class of interest [44, 45] as follows.

212 **Definition 5.1.** *A linear-memory distributed algorithm satisfies the following constraints:*

- 213 1. Local memory: *Each node i can, at time $t \geq 0$, store past values in a (finite) internal*
 214 *memory $\mathcal{M}_{i,t}$, which is updated by either the local computation or communication via*
 215 $\mathcal{M}_{i,t} \subseteq \mathcal{M}_{i,t}^{\text{comp}} \cup \mathcal{M}_{i,t}^{\text{comm}}$.
- 216 2. Local computation: *Each node i can draw a random seed r_i at the beginning, and at time t ,*
 217 *samples a random index $\xi_{i,t} \sim \Xi_i$ and then uses a randomized mapping $A_{i,t}$ to calculate the*
 218 *computation memory $\mathcal{M}_{i,t}^{\text{comp}}$ via $\mathcal{M}_{i,t}^{\text{comp}} = A_{i,t}(r_i, F_i(\mathbf{x}_{i,0}, \xi_{i,0}), \dots, F_i(\mathbf{x}_{i,t-1}, \xi_{i,t-1}))$*
 219 *and $\mathcal{M}_{i,t}^{\text{comp}} = A_{i,t}(r_i, \partial F_i(\mathbf{x}_{i,0}, \xi_{i,0}), \dots, \partial F_i(\mathbf{x}_{i,t-1}, \xi_{i,t-1}))$ for zeroth- and first-order*
 220 *methods, respectively.*
- 221 3. Local communication: *Each node i can, at time t , share a value to all or part of its neighbors*
 222 $\mathcal{N}(i)$, *which forms the communication memory via $\mathcal{M}_{i,t}^{\text{comm}} = \cup_{j \in \mathcal{N}(i)} \mathcal{M}_{j,t}$.*
- 223 4. Output value: *Each node i can, at time t , specify one vector $\mathbf{x}_{i,t} \in \mathcal{M}_{i,t}$ as the local output.*

224 The proposed NOG method naturally fits into the aforementioned distributed algorithm class, with
 225 its depth equivalent to the communication complexity. In the distributed setting, we simply need to
 226 replace the computation of F or ∇F in Propositions 4.1 and 4.2 with the average of the corresponding
 227 oracle outputs from m machines. The correctness of the described procedure can be guaranteed by
 228 the following proposition.

229 **Proposition 5.1** (Distributed estimator). *Let $\mathbf{O}_i$ be the (δ, σ) -SFO of each local function f_i for*
 230 $i \in [m]$. *The mean estimator $\mathbf{O} = 1/m \cdot \sum_{i=1}^m \mathbf{O}_i$ is a $(\delta, \sigma/\sqrt{m})$ -SFO of the global function f .*

231 The proposition shows that the distributed estimators reduce the variance by $1/m$, which, in conjunc-
 232 tion with Theorem 4.2 and 5.2, implies that the distributed algorithm enjoys a linear speedup over its
 233 single-agent counterpart, and the total SZO or SFO computation remains unchanged. In addition, the
 234 low depth of NOG method makes it much more communication-efficient than previous ones [35, 43].
 235 Below, we summarize the communication and computation complexities of the distributed algorithms
 236 in the following theorems, and compare them with previous results in Table 2.

237 **Theorem 5.1** (Distributed zeroth-order NOG). *Under Assumption 5.1, following the procedure*
 238 *described in Theorem 4.2 and splitting the calculation of F into m nodes, we obtain a distributed*
 239 *algorithm that finds a $(2\delta, \epsilon)$ -Goldstein stationary point of f in communication of $\mathcal{O}(d^{1/3}(\gamma +$
 240 $\delta L)L^{2/3}\delta^{-1}\epsilon^{-5/3})$ and SZO computation of $\mathcal{O}(d(\gamma + \delta L)L^2\delta^{-1}\epsilon^{-3})$.*

241 **Theorem 5.2** (Distributed first-order NOG). *Under Assumption 5.1, following the procedure de-*
 242 *scribed in Theorem 4.3 and the calculation of ∇F into m nodes, we obtain a distributed algorithm that*
 243 *finds a $(2\delta, \epsilon)$ -Goldstein stationary point of f in communication of $\mathcal{O}(d^{1/3}(\gamma + \delta L)L^{2/3}\delta^{-1}\epsilon^{-5/3})$*
 244 *and SFO computation of $\mathcal{O}((\gamma + \delta L)L^2\delta^{-1}\epsilon^{-3})$.*

245 6 Applications in Quantum Optimization

246 In this section, we present the quantum implementation of our proposed NOG method and demonstrate
 247 that it also achieves improved quantum query complexities than existing methods.

Order	Method	Communication	Computation	Reference
0th	DGFM	$\mathcal{O}(d^{3/2}\delta^{-1}\epsilon^{-4})$	$\mathcal{O}(d^{3/2}\delta^{-1}\epsilon^{-4})$	[35]
	DGFM+	$\mathcal{O}(d^{3/2}\delta^{-1}\epsilon^{-3})$	$\mathcal{O}(d^{3/2}\delta^{-1}\epsilon^{-3})$	[35]
	ME-DOL	$\mathcal{O}(d\delta^{-1}\epsilon^{-3})$	$\mathcal{O}(d\delta^{-1}\epsilon^{-3})$	[43]
	NOG(ours)	$\mathcal{O}(d^{1/3}\delta^{-1}\epsilon^{-5/3})$	$\mathcal{O}(d\delta^{-1}\epsilon^{-3})$	Theorem 4.2
1st	ME-DOL	$\mathcal{O}(\delta^{-1}\epsilon^{-3})$	$\mathcal{O}(\delta^{-1}\epsilon^{-3})$	[43]
	NOG(ours)	$\mathcal{O}(d^{1/3}\delta^{-1}\epsilon^{-5/3})$	$\mathcal{O}(\delta^{-1}\epsilon^{-3})$	Theorem 4.3

Table 2: Comparison of communicational and computational complexities for **distributed** algorithms for finding (δ, ϵ) stationary points of nonsmooth nonconvex problems.

6.1 Quantum Oracles

Notations of quantum computing [40]. Let $\mathcal{L}_2$ be the space of complex-valued functions $\psi : \mathbb{R}^d \rightarrow \mathbb{C}$ such that $\int_{\mathbb{R}^d} |\psi(\mathbf{x})|^2 d\mathbf{x} = 1$. Using Dirac notation, let $|\psi\rangle$ represent the wave function $\psi(\mathbf{x})$ and $\langle\psi|$ its conjugate $\overline{\psi(\mathbf{x})}$, with the inner product defined as $\langle\psi|\psi'\rangle = \int_{\mathbb{R}^d} \overline{\psi(\mathbf{x})}\psi'(\mathbf{x})d\mathbf{x}$. The probability of finding $\mathbf{x}$ in region R is $\int_R |\psi(\mathbf{x})|^2 d\mathbf{x}$. Thus, the state $|\psi\rangle$ encodes the distribution $p(\mathbf{x}) = |\psi(\mathbf{x})|^2$, where measurement yields a sample from $p(\mathbf{x})$.

An important fact of quantum computing is the following.

Fact 6.1 ([40, Section 4]). *For any function $f(\mathbf{x}) : \{0, 1\}^{n_1} \mapsto \{0, 1\}^{n_2}$ implemented by a classical circuit C , we can implement a quantum circuit C' with the same size up to constants which implements the transformation U_f such that $U_f : |\mathbf{x}\rangle \otimes |\mathbf{y}\rangle \mapsto |\mathbf{x}\rangle \otimes |f(\mathbf{x}) \oplus \mathbf{y}\rangle$.*

Based on Fact 6.1, it is natural to define the quantum stochastic zeroth-order and first-order oracles for the objective function f , which will have the same circuit complexity as their classical counterparts.

Definition 6.1 (Quantum stochastic zeroth-order oracle). *For function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, its quantum stochastic zeroth-order (QSZO) oracle is defined as $\mathbf{O}_{\text{QSZO}} : |\mathbf{x}\rangle \otimes |\xi\rangle \otimes |\mathbf{y}\rangle \mapsto |\mathbf{x}\rangle \otimes |\xi\rangle \otimes |F(\mathbf{x}, \xi) + \mathbf{y}\rangle$, where $F(\mathbf{x}, \xi)$ is the stochastic component of Problem (1).*

Definition 6.2 (Quantum stochastic first-order oracle). *For function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, its quantum stochastic first-order (QSFO) oracle is defined as $\mathbf{O}_{\text{QSFO}} : |\mathbf{x}\rangle \otimes |\xi\rangle \otimes |\mathbf{y}\rangle \mapsto |\mathbf{x}\rangle \otimes |\xi\rangle \otimes |\nabla F(\mathbf{x}, \xi) + \mathbf{y}\rangle$, where $F(\mathbf{x}, \xi)$ is the stochastic component of Problem (1).*

In a classical computer, achieving $\text{Var}[\hat{\mu}] \leq \sigma^2$ for a random variable Z requires $n = \mathcal{O}(\text{Var}[Z]/\sigma^2)$ i.i.d. samples. To obtain a quantum speedup, Sidford and Zhang [47] combined multi-level Monte Carlo (MLMC) techniques with quantum multivariate mean estimators [10]. Their resulting subroutine, `QuantumMeanEstimation+`, produces an unbiased estimator satisfying $\text{Var}[\hat{\mu}] \leq \sigma^2$ using only $\tilde{\mathcal{O}}(\sqrt{d}\text{Var}[Z]/\sigma)$ quantum sample oracle defined as follows.

Definition 6.3 (Quantum sampling oracle). *For a random variable $Z \in \mathbb{R}^d$ whose probability density function is $p_Z(\mathbf{z})$, its quantum sampling oracle $\mathbf{O}_Z$ is defined as $\mathbf{O}_Z : |\mathbf{0}\rangle \mapsto \int_{\mathbf{z} \in \mathbb{R}^d} \sqrt{p_Z(\mathbf{z})} d\mathbf{z} |\mathbf{z}\rangle \otimes |\text{garbage}(\mathbf{z})\rangle$, where $|\text{garbage}(\mathbf{z})\rangle$ denotes possible garbage states that arise during the implementation of a quantum oracle.*

Proposition 6.1 ([47, Theorem 4]). *For a random variable $Z \in \mathbb{R}^d$ with bounded variance $\text{Var}[Z] < +\infty$, there exists a quantum algorithm `QuantumMeanEstimation+` that can output an unbiased estimator $\hat{\mu}$ of $\mu = \mathbb{E}[Z]$ satisfying $\text{Var}[\hat{\mu}] \leq \sigma^2$ using $\tilde{\mathcal{O}}(\sqrt{d}\text{Var}[Z]/\sigma)$ quantum sampling oracle in expectation, where the notation $\tilde{\mathcal{O}}(\cdot)$ hides logarithmic factors of $\text{Var}[Z]/\sigma$.*

6.2 Faster Quantum Zeroth- and First-Order Methods

Based on Proposition 6.1, we can give the quantum counterpart of Proposition 4.1 as follows.

Proposition 6.2. *Under Assumption 3.1, there exists an algorithm that can construct the (δ, σ) -SFO within $\tilde{\mathcal{O}}(dL/\sigma)$ QSZO complexity in expectation.*

Then, by integrating this estimator into our proposed NOG method, we obtain the following complexity bound in the quantum zeroth-order setting, which improves the $\mathcal{O}(d^{3/2}\delta^{-1}\epsilon^{-7/3})$ QSZO complexity achieved by the recently proposed QGFM+ [36] by a factor of $d^{1/3}$.

Theorem 6.1 (Quantum zeroth-order NOG). *Under Assumption 3.1, Algorithm 1 with the parameters given by Theorem 4.1 and the (δ, σ) -SFO given by Proposition 6.2 with $\sigma = d^{1/6}L^{1/3}\epsilon^{2/3}$ can find a $(2\delta, \epsilon)$ -Goldstein stationary point of f in $\tilde{\mathcal{O}}(d^{7/6}(\gamma + \delta L)L^{4/3}\delta^{-1}\epsilon^{-7/3})$ QSZO complexity.*

Next, we consider the first-order setup. Similar to the zeroth-order case, we first provide a construction for the (δ, σ) -SFO with a quantum counterpart of Proposition 4.2.

Proposition 6.3. *Under Assumption 3.1, there exists an algorithm that can construct the (δ, σ) -SFO within $\tilde{\mathcal{O}}(\sqrt{d}L/\sigma)$ QSFO complexity in expectation.*

Finally, leveraging the above QSFO-based estimator, we achieve a further improved complexity for nonsmooth nonconvex optimization, as stated in the following theorem. Compared with Theorem 6.1, the quantum first-order method improves the zeroth-order one by a factor of $d^{1/2}$ in query complexity.

Theorem 6.2 (Quantum first-order NOG). *Under Assumption 3.1, Algorithm 1 with the parameters given by Theorem 4.1 and the (δ, σ) -SFO given by Proposition 6.3 with $\sigma = d^{1/6}L^{1/3}\epsilon^{2/3}$ can find a $(2\delta, \epsilon)$ -Goldstein stationary point of f in $\tilde{\mathcal{O}}(d^{2/3}(\gamma + \delta L)L^{4/3}\delta^{-1}\epsilon^{-7/3})$ QSFO complexity.*

7 Conclusions and Future Works

In this work, we propose the Nonconvex Optimistic Gradient (NOG) method as a meta-algorithm for highly-parallel nonsmooth nonconvex optimization. Building upon the NOG framework, we further develop communication-efficient distributed algorithms and fast quantum algorithms.

For future research, it would be of great interest to establish lower bounds for these settings. Such results would be instrumental in examining the tightness of our proposed methods. Moreover, our distributed algorithm requires a centralized server, and it is also important to consider its decentralized extensions [8, 35, 43]. Finally, noting that the δ and ϵ dependencies of our zeroth- and first-order quantum algorithms and the one in [36] are all $\delta^{-1}\epsilon^{-7/3}$, we wonder whether there exist faster algorithms that can break this barrier.

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A Proof of Randomized Smoothing

Proposition 3.1. *If f satisfies Assumption 3.1, then its smoothed surrogate f_δ satisfies:*

- (a) $|f_\delta(\cdot) - f(\cdot)| \leq \delta L$.
- (b) $f_\delta(\cdot)$ is L -Lipschitz.
- (c) $\partial_\delta f_\delta(\cdot) \in \partial_{2\delta} f(\cdot)$.
- (d) $f_\delta(\cdot)$ is differentiable and has $\mathcal{O}(\sqrt{d}L/\delta)$ -Lipschitz gradients.

Proof. We verify each items one by one as follows.

- (a) It follows the definition of f_δ (Definition 3.4) and Jensen's inequality that

$$|f_\delta(\mathbf{x}) - f(\mathbf{x})| \leq \mathbb{E}_{\mathbf{u} \sim \text{Unif}(\mathbb{B}^d)} |f(\mathbf{x} + \delta \mathbf{u}) - f(\mathbf{x})| \leq L\delta \mathbb{E}_{\mathbf{u} \sim \text{Unif}(\mathbb{B}^d)} \|\mathbf{u}\| \leq L\delta.$$

- (b) By the definition of f_δ (Definition 3.4), the Lipschitz continuity of f and Jensen's inequality, for any $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^d$ we have that

$$\begin{aligned} |f_\delta(\mathbf{x}) - f_\delta(\mathbf{x}')| &= |\mathbb{E}_{\mathbf{u} \sim \text{Unif}(\mathbb{B}^d)} [f(\mathbf{x} + \delta \mathbf{u})] - \mathbb{E}_{\mathbf{u}' \sim \text{Unif}(\mathbb{B}^d)} [f(\mathbf{x}' + \delta \mathbf{u}')]| \\ &= |\mathbb{E}_{\mathbf{u} \sim \text{Unif}(\mathbb{B}^d)} [f(\mathbf{x} + \delta \mathbf{u})] - \mathbb{E}_{\mathbf{u} \sim \text{Unif}(\mathbb{B}^d)} [f(\mathbf{x}' + \delta \mathbf{u})]| \\ &\leq \mathbb{E}_{\mathbf{u} \sim \text{Unif}(\mathbb{B}^d)} |f(\mathbf{x} + \delta \mathbf{u}) - f(\mathbf{x}' + \delta \mathbf{u})| \leq L\|\mathbf{x} - \mathbf{x}'\|. \end{aligned}$$

- (c) See [32, Lemma 4] for a proof.

- (d) It is first proven in [51]. See also [34, Proposition 2.2] for a more complete proof.

□

B Proof of the Nonconvex Optimistic Gradient Method

Lemma 4.3. *Let $\mathbf{O}$ be a (δ, σ) -SFO of a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ that satisfies Assumption 3.1. For any vectors $\mathbf{u}_1, \dots, \mathbf{u}_K \in \mathbb{B}_D^d(\mathbf{0})$, the update in Eq. (4) ensures the regret bound of*

$$\mathbb{E}[\text{Reg}_{T,K}] \leq \frac{4KD^2}{\eta} + 15\eta KM \left(\frac{9c^2 d L^2 D^2}{2\delta^2} + \sigma^2 \right).$$

Proof. We start from Lemma 4.2 that

$$\mathbb{E}[\text{Reg}_{T,K}] \leq \frac{4KD^2}{\eta} + \frac{5\eta}{2} \sum_{t=1}^{KM} \mathbb{E} \|\mathbf{O}(\mathbf{y}_t) - \mathbf{O}(\mathbf{y}_{t-1})\|^2.$$

Applying Young's inequality on the second term, we further obtain that

$$\begin{aligned} \mathbb{E}[\text{Reg}_{T,K}] &\leq \frac{4KD^2}{\eta} + \frac{15\eta}{2} \sum_{t=1}^{KM} \mathbb{E} [\|\nabla f_\delta(\mathbf{y}_t) - \nabla f_\delta(\mathbf{y}_{t-1})\|^2 \\ &\quad + \|\mathbf{O}(\mathbf{y}_t) - \nabla f_\delta(\mathbf{y}_t)\|^2 + \|\mathbf{O}(\mathbf{y}_{t-1}) - \nabla f_\delta(\mathbf{y}_{t-1})\|^2]. \end{aligned}$$

By Proposition 3.1(d) we have $\|\nabla f_\delta(\mathbf{y}_t) - \nabla f_\delta(\mathbf{y}_{t-1})\| \leq (c\sqrt{d}L/\delta)\|\mathbf{y}_t - \mathbf{y}_{t-1}\|$, and by the definition of (δ, σ) oracle (Definition 3.5) we have $\mathbb{E} \|\mathbf{O}(\mathbf{y}) - \nabla f_\delta(\mathbf{y})\|^2 \leq \sigma^2$. Therefore, we get

$$\mathbb{E}[\text{Reg}_{T,K}] \leq \frac{4KD^2}{\eta} + \frac{15\eta c^2 d L^2}{2\delta^2} \sum_{t=1}^{KM} \mathbb{E} \|\mathbf{y}_t - \mathbf{y}_{t-1}\|^2 + 15\eta KM \sigma^2.$$

In Algorithm 2, we have $\mathbf{y}_t - \mathbf{y}_{t-1} = (\mathbf{x}_{t-1} + s_t \mathbf{\Delta}_t) - (\mathbf{x}_{t-2} + s_{t-1} \mathbf{\Delta}_{t-1}) = (1 - s_{t-1}) \mathbf{\Delta}_{t-1} + s_t \mathbf{\Delta}_t \in \mathbb{B}_{3D}^d(\mathbf{0})$. Therefore, we can further bound the right-hand side in the above inequality as

$$\mathbb{E}[\text{Reg}_{T,K}] \leq \frac{4KD^2}{\eta} + 15\eta KM \left(\frac{9c^2 d L^2 D^2}{2\delta^2} + \sigma^2 \right).$$

□

Theorem 4.1. Let $\mathbf{O}$ be a (δ, σ) -SFO of a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ that satisfies Assumption 3.1. Setting the parameters in Algorithm 1 as

$$\eta = \Theta \left(\frac{\delta}{L\sqrt{dM}} \wedge \frac{D}{\sigma\sqrt{M}} \right) \quad \text{and} \quad M = \Theta \left(\left(\frac{\sqrt{d}L\delta T}{\gamma + \delta L} \right)^{2/5} \vee \left(\frac{\sigma\delta T}{\gamma + \delta L} \right)^{2/3} \right).$$

Then the algorithm can find a $(2\delta, \epsilon)$ -Goldstein stationary point of f in iterations no more than

$$T = \mathcal{O} \left(\frac{\gamma + \delta L}{\delta} \left(\frac{d^{1/3}L^{2/3}}{\epsilon^{5/3}} + \frac{\sigma^2}{\epsilon^3} \right) \right).$$

Proof. Selecting the stepsize $\eta = \Theta \left(\frac{\delta}{L\sqrt{dM}} \wedge \frac{D}{\sigma\sqrt{M}} \right)$ to minimize the regret in Lemma 4.3, we obtain

$$\mathbb{E}[\text{Reg}_{T,K}] = \mathcal{O} \left(K \left(\frac{D^2 L \sqrt{dM}}{\delta} + D \sigma \sqrt{M} \right) \right).$$

Substituting the above regret bound into Lemma 4.1, we have

$$\mathbb{E}[\text{dist}(\partial_\delta f_\delta(\bar{\mathbf{y}}_{\text{out}}), \mathbf{0})] = \mathcal{O} \left(\frac{f_\delta(\mathbf{0}) - f_\delta^*}{KDM} + \frac{\sqrt{d}LD}{\sqrt{M}} + \frac{\sigma}{\sqrt{M}} \right).$$

Note that by Assumption 3.1 we have $f(\mathbf{0}) - f^* \leq \gamma$ and by Proposition 3.1(a) we have $|f_\delta(\cdot) - f(\cdot)| \leq \delta L$. Therefore, we have $f_\delta(\mathbf{0}) - f_\delta^* \leq \gamma + \delta L$ and

$$\mathbb{E}[\text{dist}(\partial_\delta f_\delta(\bar{\mathbf{y}}_{\text{out}}), \mathbf{0})] = \mathcal{O} \left(\frac{\gamma + \delta L}{KDM} + \frac{\sqrt{d}LD}{\delta\sqrt{M}} + \frac{\sigma}{\sqrt{M}} \right).$$

For $T = KM$ and $D = \delta/M$, we have

$$\mathbb{E}[\text{dist}(\partial_\delta f_\delta(\bar{\mathbf{y}}_{\text{out}}), \mathbf{0})] = \mathcal{O} \left(\frac{M(\gamma + \delta L)}{\delta T} + \frac{\sqrt{d}L}{M^{3/2}} + \frac{\sigma}{\sqrt{M}} \right).$$

Selecting the parameter $M = \Theta \left(\left(\frac{\sqrt{d}L\delta T}{\gamma + \delta L} \right)^{2/5} \vee \left(\frac{\sigma\delta T}{\gamma + \delta L} \right)^{2/3} \right)$ to minimize the right-hand side, we obtain

$$\mathbb{E}[\text{dist}(\partial_\delta f_\delta(\bar{\mathbf{y}}_{\text{out}}), \mathbf{0})] = \mathcal{O} \left(d^{1/5} L^{2/5} \left(\frac{\gamma + \delta L}{\delta T} \right)^{3/5} + \sigma^{2/3} \left(\frac{\gamma + \delta L}{\delta T} \right)^{1/3} \right).$$

Recalling Proposition that 3.1(d) $\partial_\delta f_\delta(\cdot) \in \partial_{2\delta} f(\cdot)$, we know that the algorithm can provably find a $(2\delta, \epsilon)$ -Goldstein stationary point in total iterations no more than

$$T = \mathcal{O} \left(\frac{\gamma + \delta L}{\delta} \left(\frac{d^{1/3}L^{2/3}}{\epsilon^{5/3}} + \frac{\sigma^2}{\epsilon^3} \right) \right).$$

□

C Proof of Classical Gradient Estimators

Proposition 4.1 (Zeroth-order estimator). Under Assumption 3.1, a (δ, σ) -SFO of function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ can be implemented by setting $\mathbf{O}(\mathbf{x}) = \frac{1}{B} \sum_{j=1}^B \mathbf{O}_j(\mathbf{x})$ with batch size $B = \mathcal{O}(dL^2/\sigma^2)$ and

$$\mathbf{O}_j(\mathbf{x}) = \frac{d}{2\delta} (F(\mathbf{x} + \delta \mathbf{v}; \xi) - F(\mathbf{x} - \delta \mathbf{v}; \xi)) \mathbf{v}, \quad \mathbf{v} \sim \text{Unif}(\mathbb{S}^{d-1}), \quad \xi \sim \Xi.$$

Proof. To prove this proposition, it suffices to show that each single-batch estimator $\mathbf{O}_j$ is a $(\delta, c\sqrt{d}L)$ -SFO of the function f , which implies that the mini-batch estimator $\mathbf{O}(\mathbf{x}) = \frac{1}{B} \sum_{j=1}^B \mathbf{O}_j(\mathbf{x})$ satisfies $\mathbb{E}[\mathbf{O}(\mathbf{x})] = \nabla f_\delta(\mathbf{x})$ and $\text{Var}[\mathbf{O}(\mathbf{x})] = \frac{1}{B} \text{Var}[\mathbf{O}_1(\mathbf{x})] \leq \sigma^2$. In the following, we prove the unbiasedness and variance bound in Definition 3.5 separately.

- 473 1. Unbiasedness. See [22, Lemma 1] as an application of generalized Stokes' theorem.
- 474 2. Variance Bound. It is first proven in [46, Lemma 10], but without explicit numerical con-
- 475 stants c . A value of the constant $c = 4\sqrt[4]{2}\sqrt{\pi}$ is given by [34, Lemma D.1].

476 □

477 **Proposition 4.2** (First-order estimator). *Under Assumption 3.1, a (δ, σ) -SFO of function $f : \mathbb{R}^d \rightarrow \mathbb{R}$*
 478 *can be implemented by setting $\mathbf{O}(\mathbf{x}) = \frac{1}{B} \sum_{j=1}^B \mathbf{O}_j(\mathbf{x})$ with batch size $B = \mathcal{O}(L^2/\sigma^2)$ and*

$$\mathbf{O}_j(\mathbf{x}) = \nabla F(\mathbf{x} + \delta \mathbf{u}; \xi), \quad \text{where } \mathbf{u} \sim \text{Unif}(\mathbb{B}^d), \quad \xi \sim \Xi.$$

479 *Proof.* Similar to the proof of Proposition 4.1, it suffices to show that each single-batch estimator
 480 $\mathbf{O}_j$ is a $(\delta, c\sqrt{d}L)$ -SFO of the function f . Below, we prove the unbiasedness and variance bound in
 481 Definition 3.5 separately. We use $\mathbb{E}[\cdot]$ to denote $\mathbb{E}_{\mathbf{u} \sim \text{Unif}(\mathbb{B}^d), \xi \sim \Xi}[\cdot]$ for simplicity of notation.

- 482 1. Unbiasedness. Since $F(\cdot, \xi)$ is Lipschitz continuous, Rademacher's Theorem guarantees
 483 that it is differentiable almost everywhere with respect to the Lebesgue measure. This
 484 ensures that $\nabla F(\mathbf{x} + \delta \mathbf{u}; \xi)$ exist for almost all $\mathbf{u} \in \text{Unif}(\mathbb{B}^d)$. Then, under the condition
 485 that $\nabla F(\mathbf{x} + \delta \mathbf{u}; \xi)$ exists, we can show that the expectation jointly in $\xi, \mathbf{u}$ and derivative
 486 are interchangeable, which implies that $\mathbb{E}[\nabla F(\mathbf{x} + \delta \mathbf{u}; \xi)] = \nabla f_\delta(\mathbf{x})$. Since we have
 487 $f_\delta(\mathbf{x}) = \mathbb{E}[F(\mathbf{x} + \delta \mathbf{u}; \xi)]$ by definition, we know that the following identity holds:

$$\begin{aligned} & \lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{f_\delta(\mathbf{x} + \mathbf{h}) - f_\delta(\mathbf{x}) - \mathbb{E}\langle \nabla F(\mathbf{x} + \delta \mathbf{u}; \xi), \mathbf{h} \rangle}{\|\mathbf{h}\|} \\ &= \lim_{\mathbf{h} \rightarrow \mathbf{0}} \mathbb{E} \left[\frac{F(\mathbf{x} + \delta \mathbf{u} + \mathbf{h}; \xi) - F(\mathbf{x} + \delta \mathbf{u}; \xi) - \langle \nabla F(\mathbf{x} + \delta \mathbf{u}; \xi), \mathbf{h} \rangle}{\|\mathbf{h}\|} \right]. \end{aligned} \quad (6)$$

488 The differentiability of $\nabla F(\cdot; \xi)$ at the point $\mathbf{x} + \delta \mathbf{u}$ implies that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{F(\mathbf{x} + \delta \mathbf{u} + \mathbf{h}; \xi) - F(\mathbf{x} + \delta \mathbf{u}; \xi) - \langle \nabla F(\mathbf{x} + \delta \mathbf{u}; \xi), \mathbf{h} \rangle}{\|\mathbf{h}\|} = 0,$$

489 while Assumption 3.1 implies that

$$\frac{|F(\mathbf{x} + \delta \mathbf{u} + \mathbf{h}; \xi) - F(\mathbf{x} + \delta \mathbf{u}; \xi)|}{\|\mathbf{h}\|} \leq L(\xi), \quad \text{and} \quad \frac{|\langle \nabla F(\mathbf{x} + \delta \mathbf{u}; \xi), \mathbf{h} \rangle|}{\|\mathbf{h}\|} \leq L(\xi).$$

490 Since $\mathbb{E}[L(\xi)] \leq L$, we can apply Lebesgue's dominated convergence theorem on (6) to
 491 obtain that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{f_\delta(\mathbf{x} + \mathbf{h}) - f_\delta(\mathbf{x}) - \mathbb{E}\langle \nabla F(\mathbf{x} + \delta \mathbf{u}; \xi), \mathbf{h} \rangle}{\|\mathbf{h}\|} = 0.$$

492 This completes the proof of $\mathbb{E}[\nabla F(\mathbf{x} + \delta \mathbf{u}; \xi)] = \nabla f_\delta(\mathbf{x})$.

- 493 2. Variance Bound. We directly bound the variance with the second-order moment and obtain

$$\text{Var}[\nabla F(\mathbf{x} + \delta \mathbf{u}; \xi)] \leq \mathbb{E}\|\nabla F(\mathbf{x} + \delta \mathbf{u}; \xi)\|^2 \leq \mathbb{E}\|\nabla F(\mathbf{x} + \delta \mathbf{u}; \xi)\|^2 \leq \mathbb{E}[L(\xi)^2] \leq L^2.$$

494 □

495 **Proposition 5.1** (Distributed estimator). *Let $\mathbf{O}_i$ be the (δ, σ) -SFO of each local function f_i for*
 496 *$i \in [m]$. The mean estimator $\mathbf{O} = 1/m \cdot \sum_{i=1}^m \mathbf{O}_i$ is a $(\delta, \sigma/\sqrt{m})$ -SFO of the global function f .*

497 *Proof.* As the proofs of the single-agent estimator, we complete the proof by verifying the un-
 498 biasedness and variance bound in Definition 3.5 separately. Below, we use $\mathbb{E}[\cdot]$ to denote
 499 $\mathbb{E}_{\mathbf{u} \sim \text{Unif}(\mathbb{B}^d), \xi_1 \sim \Xi_1, \dots, \xi_m \sim \Xi_m}[\cdot]$ for the simplicity of notation.

500 1. Unbiasedness. It follows from the interchangeability of mean, derivative, and expectation:

$$\begin{aligned}\mathbb{E}[\mathbf{O}(\mathbf{x})] &= \frac{1}{m} \sum_{i=1}^m \mathbb{E}[\mathbf{O}_i(\mathbf{x})] = \frac{1}{m} \sum_{i=1}^m \nabla \mathbb{E}[f_i(\mathbf{x} + \delta \mathbf{u})] \\ &= \mathbb{E} \left[\frac{1}{m} \sum_{i=1}^m \nabla f_i(\mathbf{x} + \delta \mathbf{u}) \right] = \nabla \mathbb{E}[f(\mathbf{x} + \delta \mathbf{u})] = \nabla f_\delta(\mathbf{x}).\end{aligned}$$

501 2. Variance bound. Using the identity $\mathbb{E} \|\sum_{i=1}^m \mathbf{a}_i\|^2 = \sum_{i=1}^m \mathbb{E} \|\mathbf{a}_i\|^2$, we have

$$\text{Var} \left[\frac{1}{m} \sum_{i=1}^m \mathbf{O}_i(\mathbf{x}) \right] = \frac{1}{m^2} \sum_{i=1}^m \text{Var}[\mathbf{O}_i(\mathbf{x})] \leq \frac{\sigma^2}{m}.$$

502

□

503 D Proof of Quantum Gradient Estimators

504 We use Proposition 6.1 to both prove Proposition 6.2 and 6.3 in a similar manner. Moreover, in
505 the following proofs, we also give explicit constructions of the quantum gradient estimators for our
506 algorithms.

507 **Proposition 6.2.** *Under Assumption 3.1, there exists an algorithm that can construct the (δ, σ) -SFO*
508 *within $\tilde{O}(dL/\sigma)$ QSZO complexity in expectation.*

509 *Proof.* To invoke Proposition 6.1, we define the random variable Z_x as

$$Z_x = \frac{d}{2\delta} (F(\mathbf{x} + \delta \mathbf{v}; \xi) - F(\mathbf{x} - \delta \mathbf{v}; \xi)) \mathbf{v}, \quad \text{where } \mathbf{v} \sim \text{Unif}(\mathbb{S}^{d-1}), \xi \sim \Xi.$$

510 And in the proof of Proposition 4.1, we have shown that $\mathbb{E}[Z_x] = \nabla f_\delta(\mathbf{x})$ and $\text{Var}[Z_x] \leq 16\sqrt{2\pi}dL^2$.
511 Then by Proposition 6.1 the QuantumMeanEstimation+ can output a (δ, σ) -SFO oracle with
512 $\tilde{O}(dL/\sigma)$ quantum sampling oracle of the random variable Z_x . Finally, we finish the proof by
513 explicitly constructing the quantum sampling oracle. It is clear that it can be implemented by QSZO
514 oracle (Definition 6.1) given we have access to the quantum sampling oracle to $\mathbf{v}$ and ξ . The quantum
515 sampling oracle to $\xi \sim \Xi$ is nothing else but a uniform distribution $\int_0^1 |\xi\rangle d\xi$, which can be prepared
516 using Hadamard gates. To construct the quantum sampling oracle to $\mathbf{v} \sim \text{Unif}(\mathbb{B}^{d-1})$, we can mimic
517 the Box-Muller transform in quantum computers, such as [6, Section 5.3]. Specifically, we can first
518 prepare two uniformly distributed random variables $\int_0^1 |\xi_1\rangle d\xi_1$ and $\int_0^1 |\xi_2\rangle d\xi_2$. Then applying the
519 quantum Box-Muller transform given by

$$|\xi_1\rangle \otimes |\xi_2\rangle \mapsto \left| \sqrt{-4 \ln \xi_1} \cos 2\pi \xi_2 \right\rangle \otimes \left| \sqrt{-4 \ln \xi_1} \sin 2\pi \xi_2 \right\rangle$$

520 yields the following result:

$$\int_0^1 |\xi_1\rangle d\xi_1 \otimes \int_0^1 |\xi_2\rangle d\xi_2 \mapsto \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi}} \exp(-z_1^2/4) |z_1\rangle dz_1 \otimes \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi}} \exp(-z_2^2/4) |z_2\rangle dz_2.$$

521 By applying the above procedure on $d/2$ pairs, we can obtain the quantum sampling oracle for
522 d -dimensional normal distribution

$$\int_{\mathbb{R}^d} \frac{1}{\sqrt{4\pi}} \exp(-\|\mathbf{z}\|^2/4) |\mathbf{z}\rangle d\mathbf{z}.$$

523 Next, we can apply a normalization transform $|\mathbf{z}\rangle \mapsto |\mathbf{z}/\|\mathbf{z}\|\rangle$ to the above state, by the isotropy of
524 d -dimensional normal distribution, we get the uniform distribution over the unit sphere

$$\sqrt{\frac{2\pi^{d/2}}{\Gamma(d/2)}} \int_{\mathbb{S}^{d-1}} |\mathbf{v}\rangle d\mathbf{v}. \quad (7)$$

525

□

526 **Proposition 6.3.** *Under Assumption 3.1, there exists an algorithm that can construct the (δ, σ) -SFO*
 527 *within $\tilde{O}(\sqrt{dL}/\sigma)$ QSFO complexity in expectation.*

528 To invoke Proposition 6.1, we define the random variable Z_x as

$$Z_x = \nabla F(x + \delta \mathbf{u}; \xi), \quad \text{where } \mathbf{u} \sim \text{Unif}(\mathbb{B}^d), \quad \xi \sim \Xi.$$

529 And in the proof of Proposition 4.2, we have shown that $\mathbb{E}[Z_x] = \nabla f_\delta(x)$ and $\text{Var}[Z_x] \leq L^2$. Then
 530 by Proposition 6.1 the QuantumMeanEstimation+ can output a (δ, σ) -SFO oracle with $\tilde{O}(\sqrt{dL}/\sigma)$
 531 quantum sampling oracle of the random variable Z_x . Finally, we finish the proof by explicitly
 532 constructing the quantum sampling oracle, as we have done in the proof of Proposition 6.2. It is clear
 533 that it can be implemented by QSFO oracle (Definition 6.2) given we have access to the quantum
 534 sampling oracle to $\mathbf{u}$ and ξ . The quantum sampling oracle to $\xi \sim \Xi$ is just the same, and we only
 535 need to focus on generating the quantum sampling oracle to $\mathbf{u} \sim \text{Unif}(\mathbb{B}^d)$. We achieve this by
 536 applying the transform $|\mathbf{z}\rangle \otimes |r\rangle \mapsto |r\mathbf{z}\rangle \otimes |r\rangle$ to the uniform distribution over the unit sphere in
 537 (constructed as the same way we get Eq. 7) and the uniform distribution over $[0, 1]$ (constructed via
 538 Hadamard transform), which yields that

$$\sqrt{\frac{2\pi^{d/2}}{\Gamma(d/2)}} \int_{\mathbb{S}^{d-1}} |\mathbf{v}\rangle d\mathbf{v} \otimes \int_0^1 |r\rangle dr \mapsto \sqrt{\frac{\pi^{d/2}}{\Gamma(d/2+1)}} \int_{\mathbb{B}^d} |\mathbf{u}\rangle d\mathbf{u} \otimes \int_0^1 |r\rangle dr.$$

539 E Proof of the Complexity of Quantum Algorithms

540 **Theorem 6.1** (Quantum zeroth-order NOG). *Under Assumption 3.1, Algorithm 1 with the parameters*
 541 *given by Theorem 4.1 and the (δ, σ) -SFO given by Proposition 6.2 with $\sigma = d^{1/6}L^{1/3}\epsilon^{2/3}$ can find a*
 542 *$(2\delta, \epsilon)$ -Goldstein stationary point of f in $\tilde{O}(d^{7/6}(\gamma + \delta L)L^{4/3}\delta^{-1}\epsilon^{-7/3})$ QSZO complexity.*

543 *Proof.* Combining Theorem 4.1 and Proposition 6.2, the total QSZO complexity to find a (δ, ϵ) -
 544 Goldstein stationary point is

$$\tilde{O}\left(\frac{\gamma + \delta L}{\delta} \left(\frac{d^{1/3}L^{2/3}}{\epsilon^{5/3}} + \frac{\sigma^2}{\epsilon^3}\right) \frac{dL}{\sigma}\right).$$

545 By selecting $\sigma = d^{1/6}L^{1/3}\epsilon^{2/3}$ to minimize the complexity above, we obtain the total QSZO
 546 complexity of $\tilde{O}(d^{7/6}(\gamma + \delta L)L^{4/3}\delta^{-1}\epsilon^{-7/3})$. $\square$

547 **Theorem 6.2** (Quantum first-order NOG). *Under Assumption 3.1, Algorithm 1 with the parameters*
 548 *given by Theorem 4.1 and the (δ, σ) -SFO given by Proposition 6.3 with $\sigma = d^{1/6}L^{1/3}\epsilon^{2/3}$ can find a*
 549 *$(2\delta, \epsilon)$ -Goldstein stationary point of f in $\tilde{O}(d^{2/3}(\gamma + \delta L)L^{4/3}\delta^{-1}\epsilon^{-7/3})$ QSFO complexity.*

550 *Proof.* Combining Theorem 4.1 and Proposition 6.3, the total QSFO complexity to find a (δ, ϵ) -
 551 Goldstein stationary point is

$$\tilde{O}\left(\frac{\gamma + \delta L}{\delta} \left(\frac{d^{1/3}L^{2/3}}{\epsilon^{5/3}} + \frac{\sigma^2}{\epsilon^3}\right) \frac{\sqrt{dL}}{\sigma}\right).$$

552 By selecting $\sigma = d^{1/6}L^{1/3}\epsilon^{2/3}$ to minimize the complexity above, we obtain the total QSFO
 553 complexity of $\tilde{O}(d^{2/3}(\gamma + \delta L)L^{4/3}\delta^{-1}\epsilon^{-7/3})$. $\square$

554 F Numerical Experiments

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